

PHYS 143 – Mechanics and Thermal Physics

Gravitation

*Why do planets, the Moon and satellites move as they do,
and which one force law and one energy explain them all?*

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By the end of this week, you should be able to:

1. Apply Kepler's three laws, including the ellipse, the eccentricity and the speeds at perihelion and aphelion.
2. Calculate with Newton's law of gravitation (vector form, superposition), g above and below the surface, and weight against gravity.
3. Analyse circular orbits: v , T and the mass of the central body; geostationary orbits.
4. Use $U = -GMm/r$ for energy conservation, escape speed and the energy of a satellite.

The detailed outcomes are in the checklist at the end of the lecture.

Three contacts:

1 DESCRIBE

How do the planets move?

2 MODEL

What force produces these orbits, and how do we calculate with it?

3 CONSERVE AND APPLY

What does energy tell us about launches, escape and unfamiliar orbits?

Reading: Young and Freedman, Chapter 13, §13.1–13.7.

What you bring, and what to master

Reused, not re-taught

Week 1: vector addition, components.

Weeks 2 and 3: $a = v^2/r$ and

$\sum F_{\text{radial}} = mv^2/r$ (circular dynamics, Toolkit 7).

Week 4: work by a varying force.

Week 5: potential energy, reference level, $K + U$ conserved.

Core skills you should master

- use a , e , $r_p = a(1 - e)$, $r_a = a(1 + e)$;
- compare periods with $T^2 \propto a^3$;
- write $\vec{F}$ with a direction and add forces;
- measure r from the centre;
- set $GMm/r^2 = mv^2/r$ for a circular orbit;
- use $U = -GMm/r$ with $K + U$ conserved.

Supporting tools

The ellipse geometry, the shell theorem, vis-viva and the derivations are in the supplement; slides marked *supporting tool* stay for reference.

Contact 1: Describe

How do the planets move?

How we solve every problem this semester

IDENTIFY

What is being asked?
What quantities are known?
What physics is relevant?

SET UP

Draw a diagram.
Choose coordinates.
Predict the approximate size and direction.
Select the equations.

EXECUTE

Perform the calculation.
Carry units.
Keep track of signs.

EVALUATE

Are the units correct?
Is the magnitude reasonable?
Does the direction agree with the sketch?
Does it make physical sense?

This week the SET UP step also asks: **where is the centre, and what is r ?** Every gravity result depends on a distance measured from a centre.

But what if... we first need observations to explain. Mars is 1.52 times as far from the Sun as Earth: how long is its year, what is the shape of its path, and does it move at a steady speed?

An opening problem: the planets

Opening problem

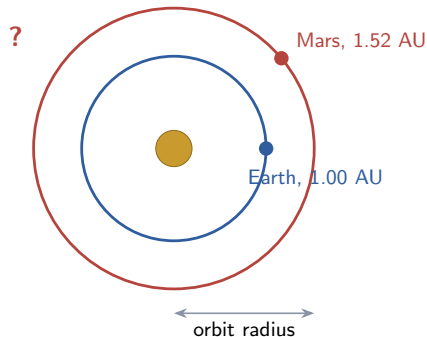
Mars orbits the Sun at 1.52 AU (1 AU is the Earth–Sun distance). Earth takes 1.00 year.

- (a) How long is a Martian year?
- (b) What is the shape of the path?
- (c) Does Mars move at a steady speed?

Sketch first, then predict

Is the Martian year shorter or longer than one Earth year? Is it 1.52 years?

But what if... we cannot answer without data. How far, and how long, for each planet?



We solve (a) with Kepler's third law, and (b) and (c) with the first two.

The solar system as data

Planet	a (AU)	T (years)	e
Mercury	0.387	0.241	0.206
Venus	0.723	0.615	0.007
Earth	1.000	1.000	0.017
Mars	1.524	1.881	0.093
Jupiter	5.203	11.86	0.048
Saturn	9.537	29.46	0.054

Units and symbols

1 AU = 1.496×10^{11} m, the mean Earth–Sun distance.

a : the **semi-major axis**, the planet's average distance (defined next).

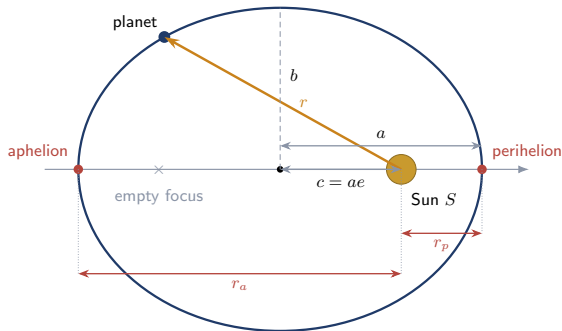
T : the orbital period.

e : the eccentricity (defined two slides on).

Interpret: farther planets take longer; but is T proportional to a ? Mars: 1.524 times as far, 1.881 times as long. No.

But what if... the data give distances, not shapes. If a planet's distance is only *on average* a , the distance must change. What path does that?

Kepler's first law: orbits are ellipses



Rule: Kepler's first law

Each planet moves in an **ellipse** with the **Sun at one focus**.

a : semi-major axis (centre to the far end).

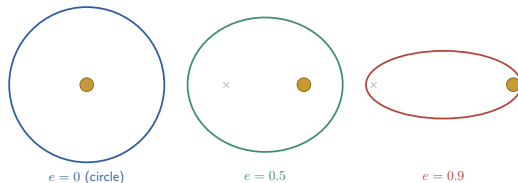
c : centre to a focus.

Perihelion: closest point, r_p .

Aphelion: farthest point, r_a .

But what if... the Sun is at a focus, not the centre. How lopsided is the ellipse? We need one number for the shape.

Eccentricity: how elongated is the ellipse?



Rule: eccentricity

$$e = \frac{c}{a}, \quad 0 \leq e < 1.$$

$e = 0$ is a circle; $e \rightarrow 1$ is a very long thin ellipse.

Rule: closest and farthest

$$r_p = a(1 - e), \quad r_a = a(1 + e)$$

$$\text{so } r_p + r_a = 2a.$$

But what if... the planets in the table have $e < 0.21$: almost circles. Comets do not. How do we find r_p and r_a in numbers?

TOOLKIT 1 – Orbit geometry

WHEN TO USE: a problem gives a , e , or the nearest and farthest distances of an elliptical orbit.

PROCEDURE: 1. **SKETCH** the ellipse, the Sun at a focus, and mark perihelion and aphelion.

2. **LIST** two of a , e , r_p , r_a ; mark the Sun's side.

3. **USE** $r_p = a(1 - e)$, $r_a = a(1 + e)$, $r_p + r_a = 2a$.

4. **CONVERT** to a and e if the other two are given.

WATCH OUT: distances are measured *from the Sun* (the focus), not from the centre of the ellipse; a is not r_p .

CHECK: $r_p < a < r_a$ and $0 \leq e < 1$; $r_p + r_a = 2a$.

Example 1: Halley's comet

Halley's comet has $a = 17.8 \text{ AU}$ and $e = 0.967$. Find r_p and r_a .

Identify: r_p and r_a , from a and e .

Set up: sketch a very long ellipse with the Sun near one end. Predict: r_p is tiny compared with a , r_a nearly $2a$.

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Execute:

$$r_p = a(1 - e) = 17.8(0.033) = 0.587 \text{ AU}$$

$$r_a = a(1 + e) = 17.8(1.967) = 35.0 \text{ AU}$$

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$$r_a = a(1 + e) = 17.8(1.967) = 35.0 \text{ AU}$$

Evaluate: $r_p + r_a = 35.6 \text{ AU} = 2a$ ✓. 0.587 AU is inside Venus's orbit; 35.0 AU is beyond Neptune, as expected for a comet.

But what if... the Sun's distance changes along the orbit. Does the planet move at the same speed all the way round?

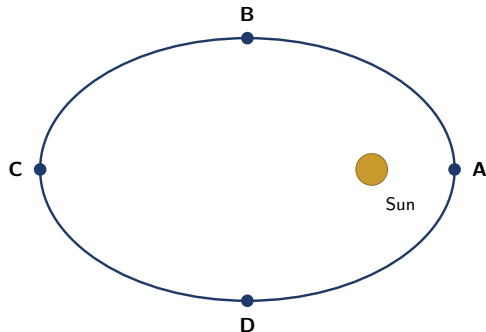
Your turn

For Earth, $a = 1.496 \times 10^{11}$ m and $e = 0.0167$. Predict whether r_p and r_a differ greatly, then calculate them.

Answer

1.471×10^{11} m and 1.521×10^{11} m:
a 3.4% difference.

Where is the planet fastest?



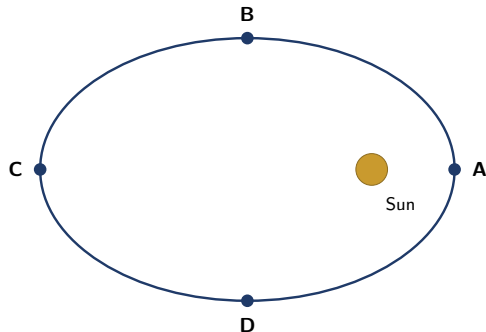
Sketch first, then predict

The planet passes A, B, C and D (A is perihelion, C is aphelion).

Where is it moving fastest? Slowest?

Does the planet's speed stay constant?

Where is the planet fastest?



Sketch first, then predict

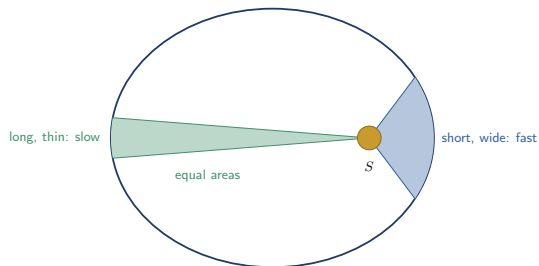
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Does the planet's speed stay constant?

Prediction guide: nearer the Sun, the Sun pulls harder and the planet speeds up, like a ball rolling down into a valley. We now state this exactly.

But what if... *faster* and *slower* are not enough. Is there a rule that tells us exactly how the speed changes along the orbit?

Kepler's second law: equal areas in equal times



Rule: Kepler's second law

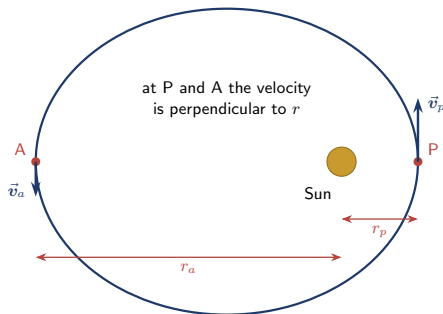
A line from the Sun to the planet sweeps out **equal areas in equal times**:

$$\frac{dA}{dt} = \text{constant}$$

The two shaded regions are swept in the *same* time. Near the Sun the region is short and wide, so the planet covers a long arc: it moves **fast**. Far from the Sun it covers a short arc: **slow**.

But what if... this tells us the speed varies, but not by how much. At the two ends of the orbit, can we turn *equal areas* into numbers?

Speeds at perihelion and aphelion



Rule: speeds at the two ends

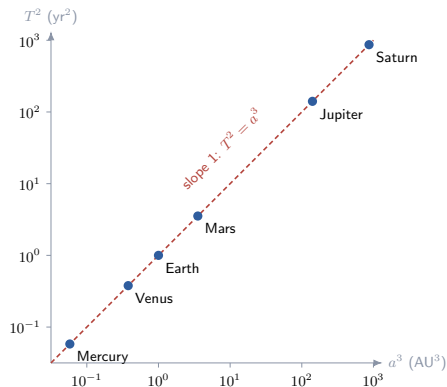
$$v_p r_p = v_a r_a$$

so $\frac{v_p}{v_a} = \frac{r_a}{r_p}$: the planet is faster at perihelion.

Why: at P and A the velocity is perpendicular to r , so in a short time Δt the swept area is a thin triangle, $\frac{1}{2} r v \Delta t$. Equal areas give $\frac{1}{2} r_p v_p \Delta t = \frac{1}{2} r_a v_a \Delta t$.

But what if... Kepler 1 and 2 describe *one* planet (the tutorial gives real speeds). What links the planets to each other, so that Mars's year follows from its distance?

Kepler's third law: periods and sizes



Rule: Kepler's third law

The square of the period is proportional to the cube of the semi-major axis:

$$T^2 \propto a^3$$

Interpret: on log axes T^2 against a^3 is a straight line of slope 1. In years and AU, $T^2/a^3 = 1.00$ for every planet in the table. So $T \propto a^{3/2}$, and a is the *semi-major axis*, not the whole major axis.

The constant is the same for all bodies orbiting the *same* central body.

But what if... we do not always have a table. Given *one* orbit, how do we get another quickly, without any constants?

TOOLKIT 2 – Ratios and scaling

WHEN TO USE: a quantity follows a power law ($T \propto a^{3/2}$, later $F \propto 1/r^2$) and a known case is given.

PROCEDURE: 1. **WRITE** the law as a ratio: $\frac{T_2}{T_1} = \left(\frac{a_2}{a_1}\right)^{3/2}$.

2. **PREDICT** bigger or smaller before calculating.

3. **SUBSTITUTE** the ratio of the *same* quantity in the same units; constants cancel.

WATCH OUT: use the ratio of the semi-major axes, not of the periods, in the bracket; the exponent is $3/2$, not $2/3$.

CHECK: units of the ratio cancel; a bigger orbit has the longer period.

Example 2: the Martian year (the opening problem)

Identify: T_{Mars} , given $a_{\text{Mars}} = 1.524 \text{ AU}$ and Earth ($a = 1 \text{ AU}$, $T = 1 \text{ yr}$).

Set up: same Sun, so Toolkit 2. Predict: longer than 1 year, but *not* 1.52 years (the law has the power $3/2$).

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Execute:

$$T = 1.00 \text{ yr} \times \left(\frac{1.524}{1.000} \right)^{3/2} = 1.88 \text{ yr}$$

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Evaluate: $1.88 > 1.52$, as predicted; $1.88 \times 365 \approx 687$ days. **The opening problem is answered:** (a) about 1.9 years; (b) an ellipse with $e = 0.093$ (Kepler 1); (c) no, faster at perihelion (Kepler 2).

Your turn

Europa and Io orbit Jupiter. Io:
 $a = 4.217 \times 10^5 \text{ km}$, $T = 1.769 \text{ days}$.
Europa: $a = 6.709 \times 10^5 \text{ km}$.
Predict, then find Europa's period.

Answer

$T = 1.769 (6.709/4.217)^{3/2} = 3.55$ days. (Scaling $T \propto a$ would wrongly give 2.81 days.)

But what if... the law works for Jupiter's moons too, but with a *different* constant than for the planets. Why does the constant depend on the central body?

Concept check: which statements are correct?

Decide for each statement, then compare with your neighbour

- I. The line from the Sun to a planet sweeps equal areas in equal times.
- II. A planet moves faster at aphelion than at perihelion.
- III. $T^2 \propto a^3$, where a is the semi-major axis.
- IV. The Sun is at the centre of a planet's elliptical orbit.

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Answers

I and III are correct.

II is false: the planet is *faster* at perihelion ($v_p r_p = v_a r_a$ with $r_p < r_a$).

IV is false: the Sun is at a focus.

But what if... the answer to II came from $v_p r_p = v_a r_a$. Is there a reason in the forces for this?
We have only described motion so far.

Common mistakes in Contact 1

Mistake	Correction
Putting the Sun at the centre of the ellipse.	The Sun is at a <i>focus</i> ; $r_p \neq r_a$.
Using $r_p = a(1 - e)$ with a as the <i>major axis</i> .	a is the <i>semi-major axis</i> , half the length.
Writing $T \propto a^2$ or $T \propto a^{2/3}$.	$T^2 \propto a^3$, so $T \propto a^{3/2}$.
Saying a planet is slowest at perihelion.	Fastest at perihelion, slowest at aphelion.
Thinking Earth's seasons come from its distance from the Sun.	They come from the tilt of the axis.
Using T in years but a in metres in a ratio.	Keep the same units for the same quantity in a ratio.

But what if... Kepler found *how* the planets move, with no reason why. What force, acting at a distance, makes $T^2 \propto a^3$? → Contact 2.

Contact 2: Model

*What force produces these orbits,
and how do we calculate with it?*

So far (Contact 1)

Kepler's laws **describe** orbits:

- ellipses, with $r_p = a(1 - e)$ and $r_a = a(1 + e)$;
- equal areas in equal times;
- $T^2 \propto a^3$.

They say nothing about *why*, or about satellites and falling objects.

We now ask

What **force** acts between the Sun and a planet, or the Earth and a satellite?

How does it change with distance and with the masses?

How do we calculate it for several bodies, and use it near and far from the surface?

Contact 2 in one line

physical idea (a force from Kepler's third law) $\rightarrow$ model (Newton's law, vector form) $\rightarrow$ method (superposition, $g(r)$, circular orbits).

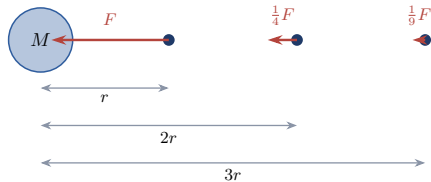
From Kepler's third law to an inverse-square force

Take a planet of mass m in a **circular** orbit of radius r (the $e \approx 0$ case), with $T^2 = k r^3$ (k constant):

$$\begin{aligned} v &= \frac{2\pi r}{T} \quad \Rightarrow \quad a_{\text{rad}} = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2} \\ &= \frac{4\pi^2 r}{k r^3} = \frac{4\pi^2}{k} \frac{1}{r^2} \end{aligned}$$

Newton's second law (Week 3, circular dynamics) gives the net force toward the Sun:

$$F = m a_{\text{rad}} = \frac{4\pi^2 m}{k} \frac{1}{r^2}$$



Rule: what Kepler 3 implies

The force on a planet falls as $\frac{1}{r^2}$:
double the distance, one quarter
the force.

But what if... $4\pi^2/k$ is a constant of the Sun, and the force is proportional to m . What is that constant, and what about the *other* body's mass?

Newton's law of universal gravitation

Rule: Newton's law of gravitation

Every particle attracts every other particle with a force directed along the line joining them, whose magnitude is

$$F = \frac{G m_1 m_2}{r^2}, \quad G = 6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2$$

Interpret

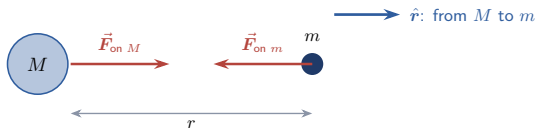
- proportional to *both* masses ($m_1 m_2$);
- inverse square of the distance r between the *centres*.

Newton's third law

The two forces are an action–reaction pair: equal in size, opposite in direction, acting on *different* bodies. The Earth pulls you with 687 N; you pull the Earth with 687 N.

But what if... the law is for particles. The Earth is not a point. Where do we measure r for a sphere?

The vector form



Rule: vector form

The force on m due to M :

$$\vec{F}_{\text{on } m} = -\frac{GMm}{r^2} \hat{r}$$

$\hat{r}$ is the unit vector pointing **from** M **to** m ; the minus sign makes the force point back toward M .

Check: $|\vec{F}| = GMm/r^2$; direction toward the other body. Swapping roles reverses $\hat{r}$, so

$$\vec{F}_{\text{on } M} = -\vec{F}_{\text{on } m}.$$

But what if... with one other body the direction is obvious. With several bodies, how do the pulls combine?

✂ TOOLKIT 3 – Superposition of gravitational forces

WHEN TO USE: a mass is attracted by two or more others (a triangle of masses, the Earth–Moon–Sun).

PROCEDURE: 1. **SKETCH** the masses and draw one arrow *toward each other body* on the mass of interest.

2. **PREDICT** the direction of the net force (between the arrows) and which arrow is longer.

3. **CALCULATE** each magnitude $F_i = Gm M_i/r_i^2$ with its *own* distance r_i .

4. **ADD** as vectors: components (Week 1, Toolkit 5), then magnitude and direction.

WATCH OUT: do not add the magnitudes unless the forces are along one line; the forces act on the same body, so there is no action–reaction cancelling.

CHECK: the net force points into the sketched region; it is smaller than the sum of the magnitudes.

Example 3: three masses

A 1.0 kg mass m is at the origin. $M_1 = 4.0$ kg is at (2.0 m, 0) and $M_2 = 6.0$ kg is at (0, 3.0 m). Find the net gravitational force on m .

Set up: $\vec{F}_1$ along $+x$, $\vec{F}_2$ along $+y$; predict the net force in the first quadrant, closer to $+x$ ($F_1 > F_2?$).

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Execute:

$$F_1 = \frac{G(1.0)(4.0)}{(2.0)^2} = 6.67 \times 10^{-11} \text{ N}$$

$$F_2 = \frac{G(1.0)(6.0)}{(3.0)^2} = 4.45 \times 10^{-11} \text{ N}$$

$$F = \sqrt{F_1^2 + F_2^2} = 8.02 \times 10^{-11} \text{ N}$$

$$\theta = \tan^{-1} \frac{F_2}{F_1} = 33.7^\circ$$

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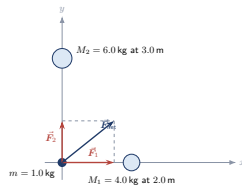
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$$F = \sqrt{F_1^2 + F_2^2} = 8.02 \times 10^{-11} \text{ N}$$

$$\theta = \tan^{-1} \frac{F_2}{F_1} = 33.7^\circ$$

Evaluate: first quadrant, nearer $+x$ ✓;
 $8.02 < 6.67 + 4.45$ ✓.



Your turn

Move M_2 to (0, -3.0 m). Predict the quadrant of $\vec{F}_{\text{net}}$, then find it.

Answer

$8.02 \times 10^{-11} \text{ N}$ at 33.7° below $+x$ (fourth quadrant).

But what if... these forces are 10^{-11} N , yet a ball near the Earth feels 9.8 N/kg . When one mass is the whole Earth, what is the simplest way to describe the pull?

Rule: gravitational acceleration at distance r from a spherical mass M

$$g(r) = \frac{GM}{r^2} \quad \text{toward the centre}$$

$$\text{For the Earth's surface: } g_s = \frac{GM_E}{R_E^2} = \frac{6.674 \times 10^{-11} (5.972 \times 10^{24})}{(6.371 \times 10^6)^2} = 9.82 \text{ m/s}^2$$

Why the mass of the object does not appear

$F = mg = GMm/r^2$: m cancels, so all objects fall with the same acceleration at one place (Week 2).

Link to Weeks 2 to 5

Near the surface $g \approx 9.8 \text{ m/s}^2$ is treated as constant: a good model while $h \ll R_E$.

But what if... g is a constant only near the ground. How does it change as we climb, and where do we measure r from?

TOOLKIT 4 – Measure r from the centre

WHEN TO USE: any gravity problem that gives a height, an altitude or a depth.

PROCEDURE: 1. **SKETCH** the body, mark its *centre*, its radius R , and the point of interest.

2. **FIND** r : above the surface $r = R + h$; below the surface $r = R - d$.

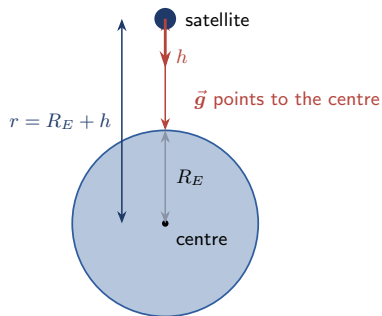
3. **CHOOSE** the formula by region: $g = GM/r^2$ for $r \geq R$; $g = g_s r/R$ for $r \leq R$ (next slides).

4. **SANITY-CHECK** with limits: $h = 0$ gives g_s ; $r \rightarrow \infty$ gives 0; the centre gives 0.

WATCH OUT: an *altitude* of 400 km is not $r = 400$ km; also never use the surface g_s at altitude.

CHECK: g at altitude is smaller than g_s , and still not zero.

g above the surface



Rule: g at altitude h

$$r = R_E + h, \quad g = g_s \left(\frac{R_E}{R_E + h} \right)^2$$

Altitude h	r/R_E	g (m/s ²)
surface, 0	1.00	9.82
400 km (ISS)	1.06	8.69
$R_E = 6371$ km	2.00	2.45
35 786 km (geostationary)	6.62	0.224

But what if... the ISS astronauts float, yet $g = 8.69 \text{ m/s}^2$ there. Is gravity absent in orbit? Let us calculate one case, then explain the floating.

Example 4: gravity at the ISS

The ISS orbits at an altitude of 400 km. Find g there.

Identify: g at $h = 4.00 \times 10^5$ m.

Set up: Toolkit 4:

$$r = R_E + h = 6.371 \times 10^6 + 0.400 \times 10^6 = 6.771 \times 10^6 \text{ m.}$$

Predict slightly less than 9.82: 6% farther from the centre, so about 12% smaller.

Example 4: gravity at the ISS

The ISS orbits at an altitude of 400 km. Find g there.

Identify: g at $h = 4.00 \times 10^5$ m.

Set up: Toolkit 4:

$$r = R_E + h = 6.371 \times 10^6 + 0.400 \times 10^6 = 6.771 \times 10^6 \text{ m.}$$

Predict slightly less than 9.82: 6% farther from the centre, so about 12% smaller.

Execute:

$$g = 9.82 \left(\frac{6.371}{6.771} \right)^2 = 9.82 (0.8853) = 8.69 \text{ m/s}^2$$

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Evaluate: $8.69 < 9.82 \checkmark$ (88.5% of g_s).

Misconception

“There is no gravity in orbit.” There is: the ISS feels 88.5% of its surface weight. Astronauts float because they are *in free fall together* with the station (weight, defined on the weight slide, reads zero).

Your turn

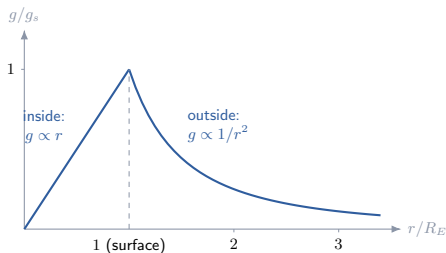
At what altitude is $g = g_s/4$?

Answer

$r = 2R_E$, so $h = R_E = 6371$ km.

But what if... this formula needs $r \geq R_E$. Dig a mine shaft: does g rise toward infinity as $r \rightarrow 0$? Surely not at the centre.

g inside the Earth (uniform-density model)



Rule: shell theorem (statement)

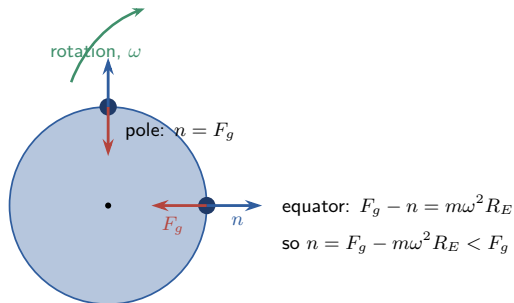
A spherical shell of mass exerts **no net force** on a body inside it. At radius $r < R$ only the mass *inside* r pulls. For uniform density, $M(r) = M (r/R)^3$, so

$$g(r) = \frac{G M(r)}{r^2} = g_s \frac{r}{R} \quad (r \leq R)$$

Interpret: g grows linearly from 0 at the centre to g_s at the surface, then falls as $1/r^2$. At a depth of 1000 km, $g = 9.82 (5371/6371) = 8.28 \text{ m/s}^2$. The real Earth has a denser core, so the uniform model gives the shape only. The proof is in the supplement.

But what if... we have the *force* at every place. A scale measures something, though. Is what a scale reads always equal to the gravitational force?

Weight against gravity



Rule: gravity and weight

Gravitational force: $F_g = GMm/r^2$, the pull of the Earth.

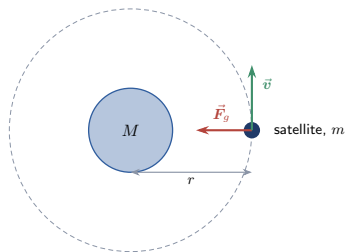
Weight (apparent weight): what a scale reads, the normal force n on the body.

At rest in an inertial frame, $n = F_g = mg$. Otherwise $n \neq F_g$.

Equator: the person moves in a circle, so $F_g - n = m\omega^2 R_E$ (Toolkit 7 of Week 3): n is about 0.35% below F_g (calculation in the tutorial). **Free fall** (a lift with a cut cable, an orbit): $n = 0$, *weightless*, although $F_g \neq 0$.

But what if... in orbit $n = 0$, yet $F_g \neq 0$: gravity alone is the inward force, so the satellite never needs an engine to keep going round. What speed and period does that require?

Circular orbits



only $\vec{F}_g$ acts:
it is the net inward force

Rule: circular orbit

Gravity is the **only** force and supplies the inward net force:

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{GM}{r}}$$

$$T = \frac{2\pi r}{v} = 2\pi \sqrt{\frac{r^3}{GM}}, \quad \omega = \sqrt{\frac{GM}{r^3}}$$

Reminder from Week 3

No extra “centripetal” or “centrifugal” arrow: $\vec{F}_g$ is the net inward force.

But what if... m cancelled: a heavy and a light satellite at the same radius?

TOOLKIT 5 – The circular-orbit method

WHEN TO USE: a satellite, moon or planet moves in a circle (or a nearly circular orbit) under gravity alone.

PROCEDURE: 1. **SKETCH** the orbit, the centre, $\vec{F}_g$ toward the centre, $\vec{v}$ tangent; mark r from the centre (Toolkit 4).

2. **WRITE** $\frac{GMm}{r^2} = \frac{mv^2}{r}$ (or $m\omega^2 r$); cancel m .

3. **SOLVE** for the unknown: $v = \sqrt{GM/r}$, or r from T , or M from v and r .

4. **CONVERT** with $v = 2\pi r/T$ to link v and T .

WATCH OUT: the satellite's mass cancels; r is $R + h$, not h ; T in seconds.

CHECK: $v \leq 7.91 \text{ km/s}$ for any Earth orbit; lower orbits are faster.

Example 5: the International Space Station

The ISS is in a circular orbit 400 km above the Earth. Find its speed and period. ($GM_E = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$.)

Identify: v and T for a circular orbit.

Set up: $r = 6.771 \times 10^6 \text{ m}$ (Toolkit 4). Predict just below the 7.91 km/s of a surface skimming orbit, and a period near 90 minutes.

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Execute:

$$v = \sqrt{\frac{3.986 \times 10^{14}}{6.771 \times 10^6}} = 7.67 \times 10^3 \text{ m/s}$$

$$T = \frac{2\pi r}{v} = \frac{2\pi(6.771 \times 10^6)}{7.67 \times 10^3} = 5.55 \times 10^3 \text{ s} = 92.4 \text{ min}$$

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Evaluate: $7.67 < 7.91 \text{ km/s}$ ✓.

But what if... this used Newton's law and the radial equation. Does it reproduce Kepler's third law, which we only observed?

Your turn (exam style)

A satellite is 200 km above the Earth. Find v for a circular orbit. Check: its radial acceleration is $g = GM/r^2$.

Answer

$r = 6.571 \times 10^6 \text{ m};$
 $v = 7.79 \text{ km/s}; a_{\text{rad}} = v^2/r =$
 $9.23 \text{ m/s}^2 = GM/r^2$ ✓.

Rule: Kepler's third law with its constant

For a circular orbit about a mass M ,

$$T^2 = \frac{4\pi^2}{GM} r^3$$

and for an ellipse the same with $r \rightarrow a$
(derivation in the supplement).

Interpret:

- $T^2 \propto a^3$ ✓, as Kepler found;
- the constant $4\pi^2/GM$ depends on the *central* body only:
Sun $2.97 \times 10^{-19} \text{ s}^2/\text{m}^3$, Earth $9.90 \times 10^{-14} \text{ s}^2/\text{m}^3$;
- the orbiting body's mass does not appear.

Concept check (exam style)

A satellite of mass m has period T in a circular orbit of radius R about the Earth. If it had mass $4m$ (same R), its period would be

- (A) $4T$ (B) $2T$ (C) T (D) $T/2$ (E) $T/4$

Kepler's third law from Newton's law

Rule: Kepler's third law with its constant

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- the orbiting body's mass does not appear.

But what if... the constant contains M . So measuring an orbit weighs the central body, with no scales at all.

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(A) $4T$ (B) $2T$ (C) T (D) $T/2$ (E) $T/4$

Answer: (C)

$T = 2\pi\sqrt{R^3/GM_E}$: m cancels.

Common mistakes in Contact 2

Mistake	Correction
Using $r = h$ (altitude) instead of $r = R + h$.	r is measured from the <i>centre</i> .
Using $g = 9.8 \text{ m/s}^2$ at orbital altitude.	g falls as $1/r^2$: 8.7 m/s^2 at 400 km.
Saying astronauts float because there is no gravity.	Gravity is still $\sim 90\%$ of g_s ; they are in free fall.
Adding forces on a mass as numbers.	Add as vectors, each toward its own source (Toolkit 3).
Adding a “centripetal” force to the diagram.	$\vec{F}_g$ is the inward net force.
Thinking a heavier satellite has a different period.	m cancels in $v = \sqrt{GM/r}$ and in T .
Using $g \propto 1/r^2$ inside the Earth.	Inside, $g \propto r$ (uniform model).

But what if... to put a satellite in orbit, someone had to raise it from the ground. $W = mgh$ would say 3.9 MJ per kg to 400 km, but g falls with height. How much energy does it really take?
→ Contact 3.

Contact 3: Conserve and apply

*What does energy tell us about
launches, escape and unfamiliar orbits?*

A new question

So far

Contact 1: Kepler's laws describe orbits.
Contact 2: $F = GMm/r^2$ explains them;
 $g(r)$ and circular orbits (v, T, M) follow.
Both use **forces**, and the force changes
with distance.

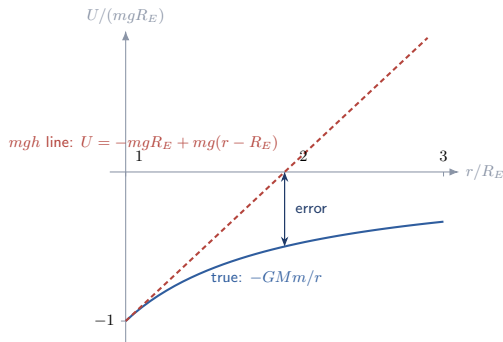
We now ask

How much **energy** does a launch, a climb
or an escape need when the force is not
constant?
What is the energy of a satellite, and
what does its sign tell us?
Can we solve problems where the method
is not given?

Contact 3 in one line

energy (Week 5) $\rightarrow U = -GMm/r \rightarrow$ escape, satellites and bound orbits $\rightarrow$ choosing the
right method for new problems.

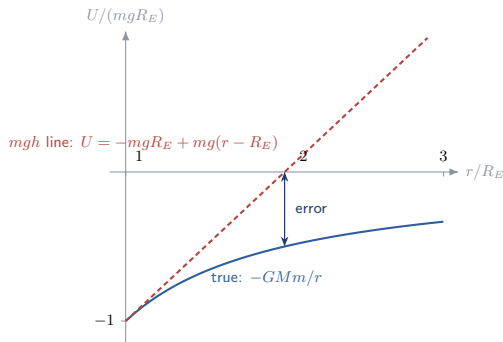
Why mgh is not enough



Sketch first, then predict

The energy to lift 1 kg from the surface to height h : is $g_s h$ exact, too large, or too small, for large h ?

Why mgh is not enough



Sketch first, then predict

The energy to lift 1 kg from the surface to height h : is $g_s h$ exact, too large, or too small, for large h ?

h	$g_s h$	true $\Delta U/m$
400 km	3.93 MJ	3.70 MJ
R_E	62.6 MJ	31.3 MJ

g falls with height, so mgh overestimates: 6% at 400 km and a factor 2 at $h = R_E$.

But what if... we need the work of a force that varies as $1/r^2$. What function $U(r)$ has $F = -dU/dr$ equal to the gravitational force?

Gravitational potential energy over large distances

Rule: gravitational potential energy

Two masses M and m , centres a distance r apart, have

$$U(r) = -\frac{GMm}{r}, \quad U \rightarrow 0 \text{ as } r \rightarrow \infty$$

The reference level ($U = 0$) is at infinite separation; U is always **negative**, rising toward zero as r increases.

Meaning of the minus sign

Gravity pulls masses together, so work must be done *against* it to separate them to infinity: a bound system has *less* energy than the separated one. Working in the supplement.

Link to mgh (Week 5)

$$\Delta U = GMm \frac{h}{R(R+h)} \approx m g_s h \text{ for } h \ll R. \text{ At } 400 \text{ km: } 3.70 \text{ against } 3.93 \text{ MJ/kg (6\% off).}$$

But what if... how do we use a negative U to find a speed or a height? $\rightarrow$ a method.

TOOLKIT 6 – The energy method for gravity

WHEN TO USE: speeds, heights or energies at two positions, when r changes by a large amount or the body leaves the Earth.

PROCEDURE: 1. **SKETCH** the two states; mark the centre and both distances r_1 and r_2 (Toolkit 4).

2. **REFERENCE:** $U = 0$ at infinity, so $U = -GMm/r$ (not mgh).

3. **WRITE** $\frac{1}{2}mv_1^2 - \frac{GMm}{r_1} = \frac{1}{2}mv_2^2 - \frac{GMm}{r_2}$ (no other forces do work).

4. **SOLVE** for the unknown; m cancels.

WATCH OUT: U is negative: signs of the terms; r measured from the centre; air and engines are not included.

CHECK: $K \geq 0$ at every position; for $h \ll R$ the answer agrees with mgh .

Example 6: how high does a probe go?

A probe is launched straight up at 8.0 km/s from the Earth's surface. Ignoring air and rotation, find the greatest height reached.

Identify: h_{max} , where $v = 0$.

Set up: Toolkit 6. Predict: more than the $v^2/2g_s = 3.3 \times 10^6 \text{ m}$ that constant- g would give, because g weakens.

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Execute:

$$\begin{aligned}\frac{1}{2}v_0^2 - \frac{GM}{R_E} &= -\frac{GM}{r_{\max}} \\ \frac{1}{r_{\max}} &= 1.570 \times 10^{-7} - 8.03 \times 10^{-8} \\ r_{\max} &= 1.304 \times 10^7 \text{ m}, \quad h = 6.67 \times 10^6 \text{ m}\end{aligned}$$

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Evaluate: $6.67 \times 10^3 \text{ km} > 3.3 \times 10^3 \text{ km}$ ✓; $v_0 < 11.2 \text{ km/s}$, so the probe returns.

But what if... at 11.0 km/s the probe goes $30R_E$ out; a little faster, does it come back at all?

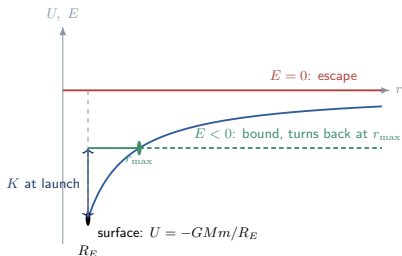
Your turn

Launched at $v_0 = 11.0 \text{ km/s}$, will the probe go higher or lower than $r = 2R_E$? Predict, then show.

Answer

Higher: $1/r_{\max} = 1.570 \times 10^{-7} - 1.518 \times 10^{-7}$, so $r_{\max} = 1.9 \times 10^8 \text{ m} = 30R_E$, far beyond $2R_E$.

Escape speed



Interpret: $E < 0$ turns back at $r_{\max}$ (Example 6). The projectile's mass does not appear; the planet's M and R do. Earth:

$$\sqrt{2(3.986 \times 10^{14})/6.371 \times 10^6} = 11.2 \text{ km/s.}$$

But what if... escape leaves with $E = 0$. A satellite in orbit has $E \neq 0$. What is its energy, and what is its sign?

Rule: escape speed

The least launch speed to reach $r \rightarrow \infty$ (arriving with $v \rightarrow 0$) has $E = 0$:

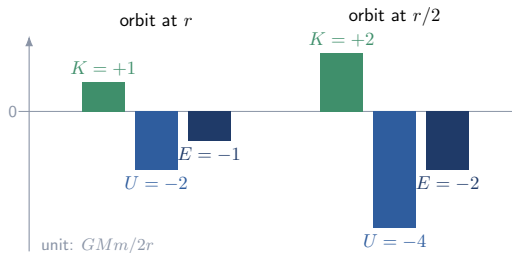
$$\frac{1}{2}mv_{\text{esc}}^2 - \frac{GMm}{R} = 0$$

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}} = \sqrt{2} v_{\text{circ}}$$

Body	Moon	Mars	Earth	Jup.	Sun
v_{esc} (km/s)	2.38	5.03	11.2	59.5	618

The Moon cannot hold an atmosphere; Lecture 10 uses 11.2 km/s for hydrogen on Earth.

The energy of a satellite



Rule: circular orbit energies

From $mv^2 = GMm/r$:

$$K = \frac{GMm}{2r}, \quad U = -\frac{GMm}{r}$$

$$E = K + U = -\frac{GMm}{2r} = -K = \frac{1}{2}U$$

Interpret: $E < 0$ means **bound**. A smaller orbit has *more negative* E and *larger* K .

But what if... air drag removes energy from a satellite. Does it slow down? E gets more negative: what happens to r and to v ?

The drag paradox

Sketch first, then predict

A low satellite loses a little energy to the thin atmosphere. After a while, is its speed larger, smaller, or the same? Does its orbit rise or fall?

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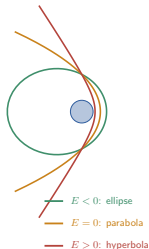
Reasoning with $E = -GMm/2r$

E decreases $\Rightarrow r$ decreases (orbit falls). Then $v = \sqrt{GM/r}$ increases: $K = -E$ rises, because U falls twice as fast as E . Drag *speeds the satellite up* as it spirals in.

Misconception

“Drag slows a satellite down.” Over the long run it makes the satellite *faster* and lower, until it burns up. To *raise* an orbit, an engine adds energy, and the satellite ends up *slower*.

But what if... these are circles. Launch it with *less* or *more* speed than the circular value: is the path still closed?



Rule: the sign of E fixes the orbit

$E < 0$: closed orbit (ellipse, circle); for an ellipse $E = -\frac{GMm}{2a}$ (semi-major axis; derivation and vis-viva in the supplement).

$E = 0$: open (parabola): just escapes.

$E > 0$: open (hyperbola): leaves with speed to spare.

Interpret: a comet with $E < 0$, like Halley's, returns; a one-time visitor from outside has $E \geq 0$. The shapes are the three conic sections; Kepler's laws describe the closed case.

But what if... ratios, forces, energy: which tool for a new problem? → Toolkit 7.

TOOLKIT 7 – Choose the method

PROCEDURE: 1. **IDENTIFY** what is asked: a force, a time, a speed, an energy, a mass, a distance.
2. **MATCH** to the tool, then **check its conditions**:

Clue in the problem	Method	Conditions
two orbits about the same body, a ratio of periods or sizes	Kepler ratio (Toolkit 2)	same central mass
net force, g at a place, forces from several bodies	Newton's law (Toolkits 3, 4)	r from centres
v , T or M for a circular orbit	circular-orbit method (Toolkit 5)	gravity only, circle
speeds or heights at two places, escape, energy to move	energy (Toolkit 6)	$U = -GMm/r$
speed at a point of an ellipse; e , r_p , r_a	Toolkit 1, $v_p r_p = v_a r_a$ (vis- viva: supplement)	closed orbit

3. **CROSS-CHECK** with a second tool when two apply (e.g. $E = -GMm/2r$ with $v = \sqrt{GM/r}$).

CHECK: units, sign of E , and the limits 7.9 km/s and 11.2 km/s.

Example 7: an integrated problem (method not given)

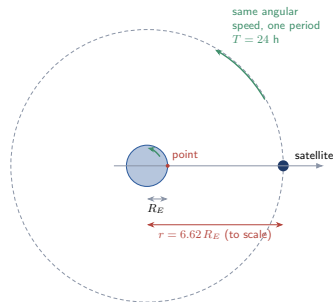
Application problem: decide what physics applies

A communications satellite stays directly above one point on the equator.

(a) At what altitude is it? (b) What is its speed? (c) Why can this happen only above the equator?

Data: the Earth turns once in 86 164 s;

$GM_E = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$; $R_E = 6.371 \times 10^6 \text{ m}$.



Sketch first, then predict

Is the orbit higher or lower than the Moon's? Will its speed be more or less than the ISS's 7.67 km/s? Which tool does the clue "the period is known" point to?

Example 7: solution

Identify: r and v for a circular orbit with $T = 86\,164\text{ s}$
(Toolkit 7: circular orbit).

Set up: the satellite must have the Earth's angular speed, so T equals the Earth's period. Gravity alone, circular.

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Identify: r and v for a circular orbit with $T = 86\,164\text{ s}$
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Execute:

$$r = \left(\frac{GM_E T^2}{4\pi^2} \right)^{1/3} = 4.216 \times 10^7 \text{ m}$$

$$h = r - R_E = 3.58 \times 10^7 \text{ m}, \quad v = \frac{2\pi r}{T} = 3.07 \times 10^3 \text{ m/s}$$

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$$h = r - R_E = 3.58 \times 10^7 \text{ m}, \quad v = \frac{2\pi r}{T} = 3.07 \times 10^3 \text{ m/s}$$

Evaluate: $h = 35\,800 \text{ km}$, well below the Moon ($3.84 \times 10^5 \text{ km}$); $v = 3.07 < 7.67 \text{ km/s}$ (higher orbits are slower) ✓. Check: $v = \sqrt{GM/r} = 3.07 \text{ km/s}$ ✓.

(c) The equator

Gravity points to the Earth's centre, so every orbit lies in a plane through the centre. Only the equatorial plane contains a circle that keeps pace with a point on the equator; any other orbit would swing north and south.

Cross-check with a ratio

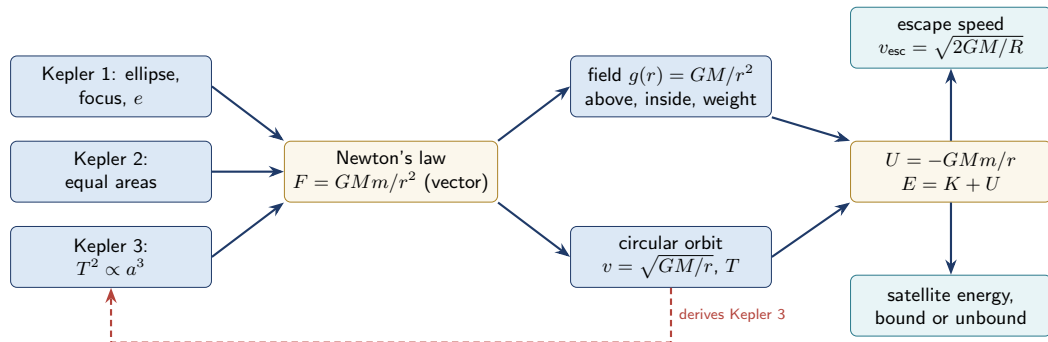
The Moon has $T = 27.3 \text{ d}$ at $3.84 \times 10^8 \text{ m}$:
 $r = 3.84 \times 10^8 (1/27.3)^{2/3} = 4.2 \times 10^7 \text{ m}$ ✓ (Toolkit 2).

Common mistakes in Contact 3

Mistake	Correction
Using $U = mgh$ for a launch from the Earth to a large height.	$U = -GMm/r$; mgh only for $h \ll R$.
Dropping the minus sign of U , or treating $E < 0$ as impossible.	$E < 0$ means bound; $K \geq 0$ is the real condition.
Using $r = \text{altitude}$ in U or v_{esc} .	r from the <i>centre</i> : $R + h$.
Thinking the escape speed depends on the mass of the projectile.	m cancels: $\sqrt{2GM/R}$ depends on the planet only.
Believing drag slows a satellite down for good.	E falls, the orbit shrinks, v rises.
Using $E = -GMm/2r$ for an elliptical orbit.	$E = -GMm/2a$, with the <i>semi-major axis</i> .
Mixing Kepler ratios of different central bodies.	The ratio T^2/a^3 is the same only for one central body.

But what if... we have three tools (ratios, forces, energy). How do they fit together, and where do we use them next? → Summary.

Week 6 concept map



Key ideas

- **Kepler's laws describe orbits:** ellipse with the Sun at a focus; equal areas in equal times; $T^2 \propto a^3$.
- **Newton's law explains them:** $F = GMm/r^2$, attractive, with r between centres; $\vec{F} = -(GMm/r^2) \hat{r}$; forces from several bodies add as vectors.
- $g = GM/r^2$ **outside**, $g = g_s r/R$ **inside (uniform model)**; weight is the scale reading, not always F_g .
- **Circular orbit:** $v = \sqrt{GM/r}$, $T = 2\pi\sqrt{r^3/GM}$; the satellite's mass cancels; $T^2 = 4\pi^2 r^3/GM$ weighs the central body.
- $U = -GMm/r$ (zero at infinity); $E = K + U$ is conserved; $v_{\text{esc}} = \sqrt{2GM/R}$; $E = -GMm/2r$ (circle), $-GMm/2a$ (ellipse); $E < 0$ means bound.
- Habit: sketch with the centre and r , predict, choose the tool, evaluate.

Reference tools: where to look

Equations, procedures: the next slides (core equations, master procedure, seven toolkits).

Definitions: the equation-anatomy slides.

Common mistakes: the last slide of each contact.

What you should remember: the chain, Contact 1

Concept	Limitation that motivates the next	Leads to
Framework, opening problem	need data: distances and periods of the planets	Solar-system data
Solar-system data	T is not proportional to a ; the distance is only an average	Kepler 1
Kepler 1 (ellipse)	the Sun is at a focus: how elongated?	Eccentricity
Eccentricity, Toolkit 1	the Sun's distance changes: is the speed constant?	Where is it fastest?
Equal areas (Kepler 2)	faster or slower is not a number: ends of the orbit	$v_p r_p = v_a r_a$
Speeds at the ends	one planet only: what links planets?	Kepler 3
Kepler 3, Toolkit 2	different constant for Jupiter's moons: why?	Concept check
Common mistakes	describes, but does not explain: what force?	Contact 2

What you should remember: the chain, Contact 2

Concept	Limitation that motivates the next	Leads to
Kepler 3 gives $1/r^2$	what constant, and the other body's mass?	Newton's law
Newton's law	for particles; where is r for a sphere?	Vector form
Vector form	one source only: several bodies?	Toolkit 3, Example 3
Superposition	10^{-11} N, yet a ball feels 9.8 N/kg	Field g
$g = GM/r^2$	constant only near the ground	g above the surface
Toolkit 4, Example 4	needs $r \geq R_E$; what below the surface?	g inside
Shell theorem	real Earth: what does a scale read?	Weight against gravity
Weight	a satellite needs an inward force only from gravity	Circular orbits
Toolkit 5, Example 5	does it reproduce Kepler 3?	Kepler 3 from Newton
Kepler 3 with its constant	the constant contains M : an orbit weighs the central body	Contact 3
Common mistakes	raising a satellite: mgh fails	Contact 3

What you should remember: the chain, Contact 3

Concept	Limitation that motivates the next	Leads to
<i>mgh</i> fails	which U gives the inverse-square force?	$U = -GMm/r$
$U = -GMm/r$	how do we use it to find a speed or height?	Toolkit 6
Toolkit 6, Example 6	does the probe ever not come back?	Escape speed
Escape speed	what is a satellite's energy?	Satellite energy
$E = -GMm/2r$	drag removes energy: slower or faster?	Drag paradox
Drag paradox	only circles so far	Bound and unbound
E fixes the orbit	many tools: which one?	Toolkit 7
Toolkit 7, Example 7	ratios, forces, energy: how they fit	Summary

Reference tools: core equations

Idea	Equation	Conditions and units
Ellipse	$r_p = a(1 - e), r_a = a(1 + e), r_p + r_a = 2a$	r from the Sun; a semi-major
Kepler 2 (ends)	$v_p r_p = v_a r_a$	velocity $\perp r$ at both ends
Kepler 3	$T^2 \propto a^3; T_2/T_1 = (a_2/a_1)^{3/2}$	same central body
Gravitation	$F = Gm_1m_2/r^2; \vec{F} = -(GMm/r^2)\hat{r}$	r between centres; N
Field	$g = GM/r^2 (r \geq R); g = g_s r/R (r \leq R)$	uniform density inside; m/s^2
Weight	$n = F_g - m\omega^2 R_E$ at the equator	scale reading, N
Circular orbit	$v = \sqrt{GM/r}, T = 2\pi\sqrt{r^3/GM}, T^2 = 4\pi^2 r^3 / GM$	gravity only; r from centre
Potential energy	$U = -GMm/r$	zero at infinity; J
Escape speed	$v_{\text{esc}} = \sqrt{2GM/R}$	$E = 0$; no air
Energies	$E = -GMm/2r$ (circle), $-GMm/2a$ (ellipse)	$E < 0$: bound
Vis-viva	$v^2 = GM(2/r - 1/a)$	closed orbit; m/s

MASTER TOOLKIT – For a gravitation problem

1. **IDENTIFY** What is asked: force, g , period, speed, energy, mass?
2. **SKETCH** Body, orbit, the forces or the two states.
3. **CENTRE AND r** Mark the centre; r from the centre ($R + h$, or $R - d$).
4. **PREDICT** Bigger or smaller than g_s , 7.9, 11.2 km/s?
5. **MODEL** Kepler ratio, Newton's law, circular orbit, or energy (Toolkit 7).
6. **EXECUTE** Algebra first, numbers last; carry units.
7. **EVALUATE** Units, sign of E , magnitude, limits ($h = 0$, $r \rightarrow \infty$).

If the result disagrees with the sketch or the prediction, investigate before accepting it.

Inside the course framework

IDENTIFY (step 1): what is being asked?

SET UP (steps 2 to 5): sketch, centre and r , predict, model.

EXECUTE (step 6): calculate.

EVALUATE (step 7): check the result.

THE SEVEN TOOLKITS

- | | | |
|----------|---|---|
| 1 | Orbit geometry | $a, e, r_p = a(1 - e), r_a = a(1 + e), r_p + r_a = 2a$; Sun at a focus |
| 2 | Ratios and scaling | $T_2/T_1 = (a_2/a_1)^{3/2}$; constants cancel; predict first |
| 3 | Superposition of forces | arrow toward each source; magnitudes; add as vectors |
| 4 | Measure r from the centre | $r = R + h$ or $R - d$; g outside and inside; limit checks |
| 5 | Circular-orbit method | $GMm/r^2 = mv^2/r$; m cancels; $T = 2\pi r/v$ |
| 6 | Energy method for gravity | $K + U$ conserved, $U = -GMm/r$; escape; satellite energies |
| 7 | Choose the method | clue $\rightarrow$ Kepler ratio, force, energy or vis-viva; cross-check |

Extension: the shell theorem, vis-viva and Kepler 2 from angular momentum are in the supplementary notes.

Common mistakes: the five that cost most marks

1. The wrong distance

r is from the *centre*: $R + h$. Using h or R alone changes g , v and U .

2. Kepler's third law

$T^2 \propto a^3$ with the *semi-major axis*, so $T \propto a^{3/2}$; and same central body.

3. Where the Sun is, and where the planet is fastest

Sun at a *focus*; fastest at perihelion;
 $r_p = a(1 - e)$.

4. Mass in the orbit formulas

The satellite's mass cancels in v and T (and in v_{esc}); the *central* mass is what matters.

5. Signs of energy

$U = -GMm/r < 0$; $E < 0$ is bound;
 $E = -GMm/2r$ (circle) or $-GMm/2a$ (ellipse).
A satellite that loses energy speeds up.

Lecture 7: momentum

Gravity as an external force; collisions and conservation laws use the same $K + U$ bookkeeping.

Lecture 8: rotation

Kepler's second law (equal areas) is conservation of angular momentum for a planet: $L = mvr$ constant gives $v_p r_p = v_a r_a$ (supplement now, derived in Lecture 8).

Lectures 10 and 11: thermal physics

The escape speed (11.2 km/s) decides which gases an atmosphere keeps: the speed distribution of molecules comes in Lecture 10.

Physical meaning

Later courses use the same inverse-square form $F \propto 1/r^2$ and $U \propto 1/r$ for the electric force; the method (sketch, centre, predict, force or energy) carries over unchanged.

Checklist: can I ...

- ☐ state Kepler's three laws and say what each is about?
- ☐ find r_p , r_a , a and e for an orbit?
- ☐ compare speeds at perihelion and aphelion?
- ☐ compare periods with $T_2/T_1 = (a_2/a_1)^{3/2}$?
- ☐ write the force between two masses, in vector form?
- ☐ add the gravitational forces from several bodies?
- ☐ find g above and below the surface, measuring r from the centre?
- ☐ tell weight from the gravitational force?
- ☐ find v and T for a circular orbit, and the mass of the central body?
- ☐ explain why the satellite's mass does not matter?
- ☐ use $U = -GMm/r$ with energy conservation?
- ☐ find the escape speed and the energy of a satellite?
- ☐ say what the sign of E tells me?
- ☐ choose between ratio, force and energy methods?
- ☐ use IDENTIFY $\rightarrow$ SET UP $\rightarrow$ EXECUTE $\rightarrow$ EVALUATE?

Next

Tutorial sheet 6: conceptual, guided, independent and application problems with answers, and an exam-style page of ten items. Supplementary notes: Kepler's second law from angular momentum, the shell theorem, $U = -GMm/r$, escape speed, $E = -GMm/2a$, vis-viva, and Kepler's third law for ellipses.